

Homework 8

40 points

1. For the given function, perform 3 steps of gradient descent. Estimate what value does x approach as the iterations continue. **2+2+2+1= 7 points**

$$f(x) = (x - 4)^2$$

2. Given $f(x) = (x - 2)^2$. Compare the results of one step of gradient descent from $x_0 = 0$ for $\alpha = 0.1$ and $\alpha = 1.5$. Which case “overshoots” the minimum? **2+2+1=5 points**

3. Evaluate the following integrals: **5*5 = 25 points**

- $\int (2x + 3)^5 dx$
- $\int \frac{1}{\sqrt{(1-x^2)}} dx$
- $\int e^x \cos x dx$
- $\int_1^4 \left(\frac{8}{\sqrt{t}} - 12\sqrt{t^3} \right) dt$
- $\int_{-4}^{-1} \left(\sqrt[3]{(5-2y)} + \frac{7}{(5-2y)} \right) dy$

4. The velocity of a particle is $v(t) = 3t^2 - 2t + 1$ m/sec. Find the displacement from $t = 0$ to $t = 4$. **3 points**